Project Overview

This project is a multi-tier Pokémon ETB Tracker application built using Python, MySQL, and an HTML/JavaScript client. The project uses a layered design with a client layer, service layer, business layer, and data layer. The application allows users to create, view, and update Pokémon ETB items stored in a MySQL database.

The Project Architecture

The project is organized into four layers:

- **Client layer:** HTML and JavaScript front end
- **Service layer:** FastAPI
- **Business layer:** Python business logic
- **Data layer:** MySQL database with DAO access methods

Prerequisites

The project is organized into four layers:

- **Client layer:** HTML and JavaScript front end
- **Service layer:** FastAPI REST API
- **Business layer:** Python business logic
- **Data layer:** MySQL database with DAO access methods

How to Download the Project

1) Go to the GitHub repository.

2) Download the ZIP file or clone the repository using Git.

3) Extract the files to a local folder.

4) Open the project folder in a terminal or code editor.

If you want, add the clone command:

git clone https://github.com/Billycolquitt12/pokemon-center-etb-tracker.git

Database setup

1) Open MySQL Workbench.
2) Open the createtables.sql file and run it.
3) Open the insertdata.sql file and run it.

4) Confirm that the tables Users, ProductSets, PokemonItems, and ItemValues were created and populated.

example verification queries:

```
USE pokemon_item_tracker;
SELECT * FROM Users;
SELECT * FROM ProductSets;
SELECT * FROM PokemonItems;
SELECT * FROM ItemValues;
```

Backend Setup

1) Open a terminal in the project folder.
2) Install the required Python packages.
3) Start the FastAPI server.
4) Commands:

```
pip install fastapi uvicorn mysql-connector-python
python -m uvicorn service.app:app –reload
```

If setup is successful, the terminal will display that Uvicorn is running on http://127.0.0.1:8000.

Frontend Setup

The front end is an HTML and JavaScript client. Open the client HTML file in a browser after the API is running. The client can be used to send requests to the FastAPI service and display returned data.

If you used a local server for the client, say:

cd client
python -m http.server 5500

Then open:

http://localhost:5500

How to Test the System

The system was tested using the HTML client and MySQL Workbench. The following functionality was tested:

- Create a new item
- Get all items
- Get a single item by ID
- Update an existing item

- Delete an item

Create

- A new Pokémon ETB item was created in the client using a valid `user_id` and `set_id`. The new row was confirmed in MySQL Workbench.

Get All

- The “Get All Items” function returned the complete list of items from the database.

Get by ID

- A specific item was retrieved by its `item_id`.

Update

- An item was updated in the client, and the changed values were verified in MySQL Workbench.

Delete

- The item was deleted from the client, and the row was confirmed removed from the database.

How to Know the Setup Worked

The setup is considered successful if:

- The API starts without errors
- Swagger opens at http://127.0.0.1:8000/docs
- The HTML client is able to retrieve data
- Database changes made through the client appear in MySQL Workbench

Screenshots on the correct steps

Figure 1 – API running

```
PS C:\Users\assau\Desktop\pokemon-center-etb-tracker> python -m uvicorn service.app:app --reload
INFO:     Will watch for changes in these directories: ['C:\\Users\\assau\\Desktop\\pokemon-center-etb-tracker']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [21236] using StatReload
INFO:     Started server process [3996]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
```

Figure 2 – Swagger documentation

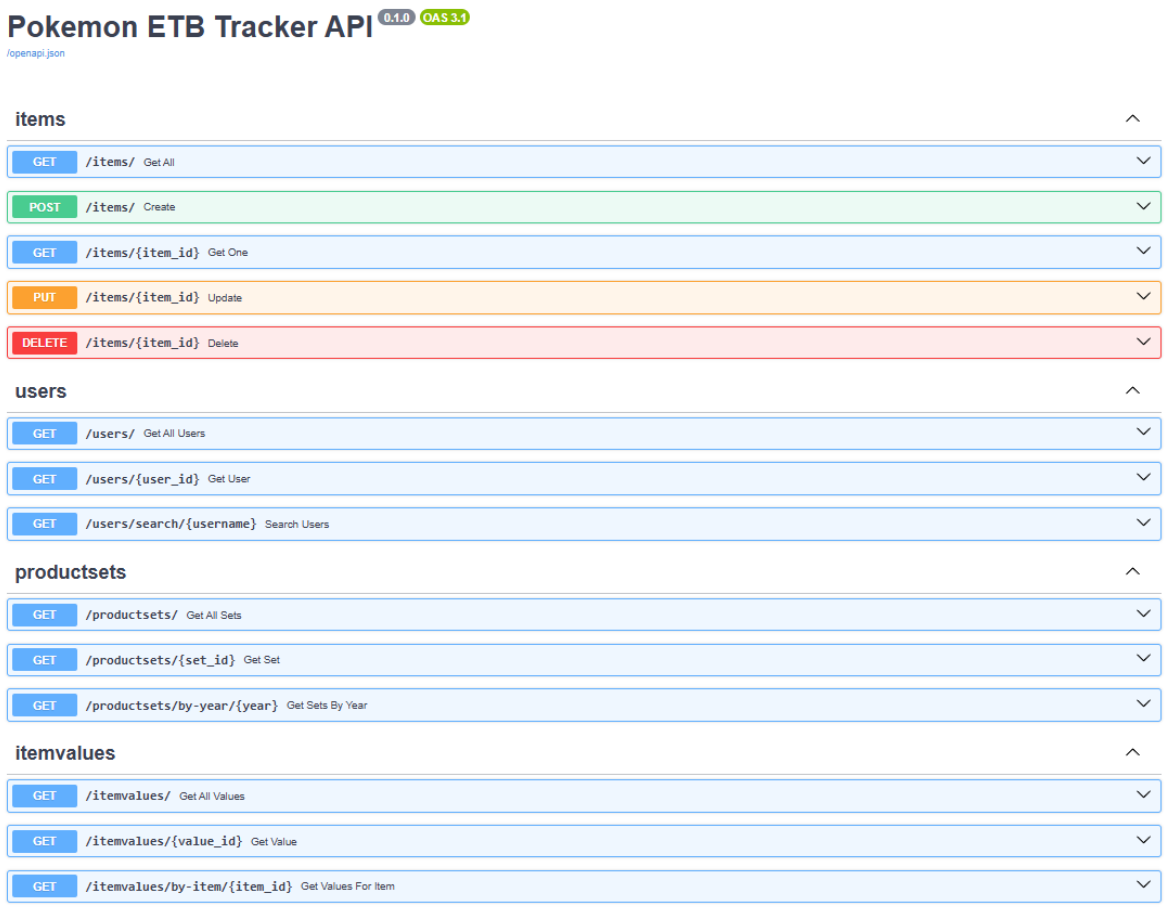

Figure 3 – Create item in client

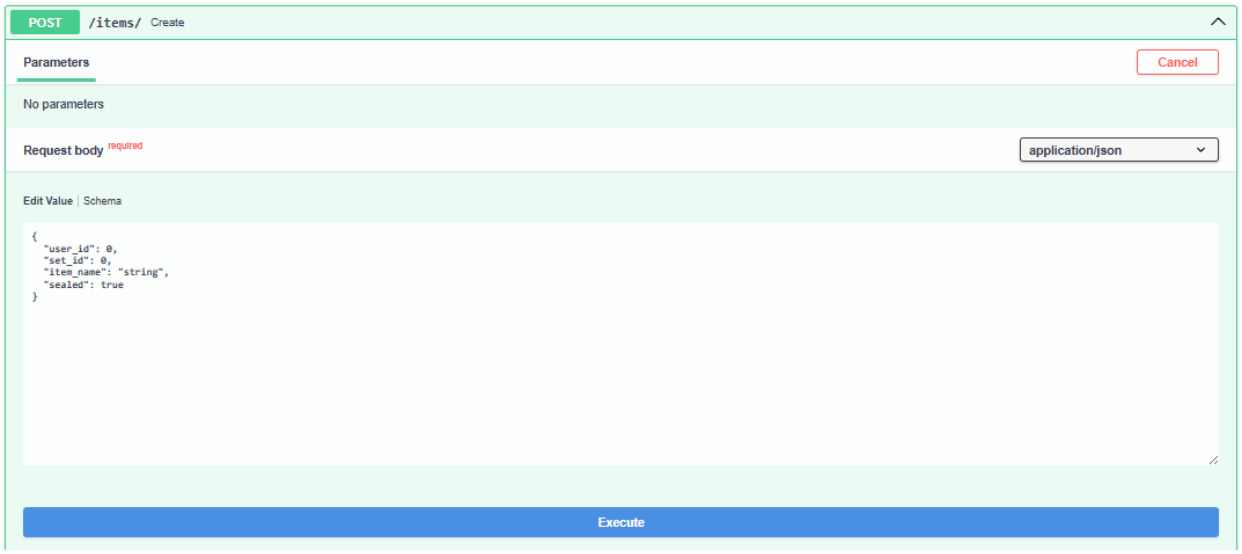

Figure 4 – Get all items

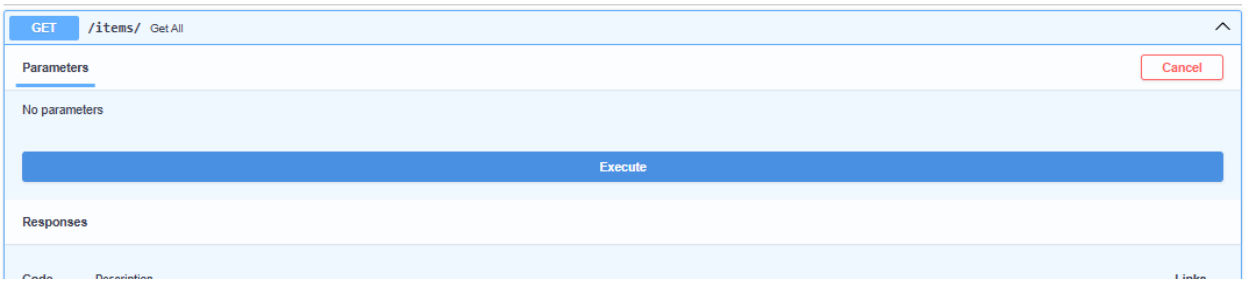

Figure 5 – Update item

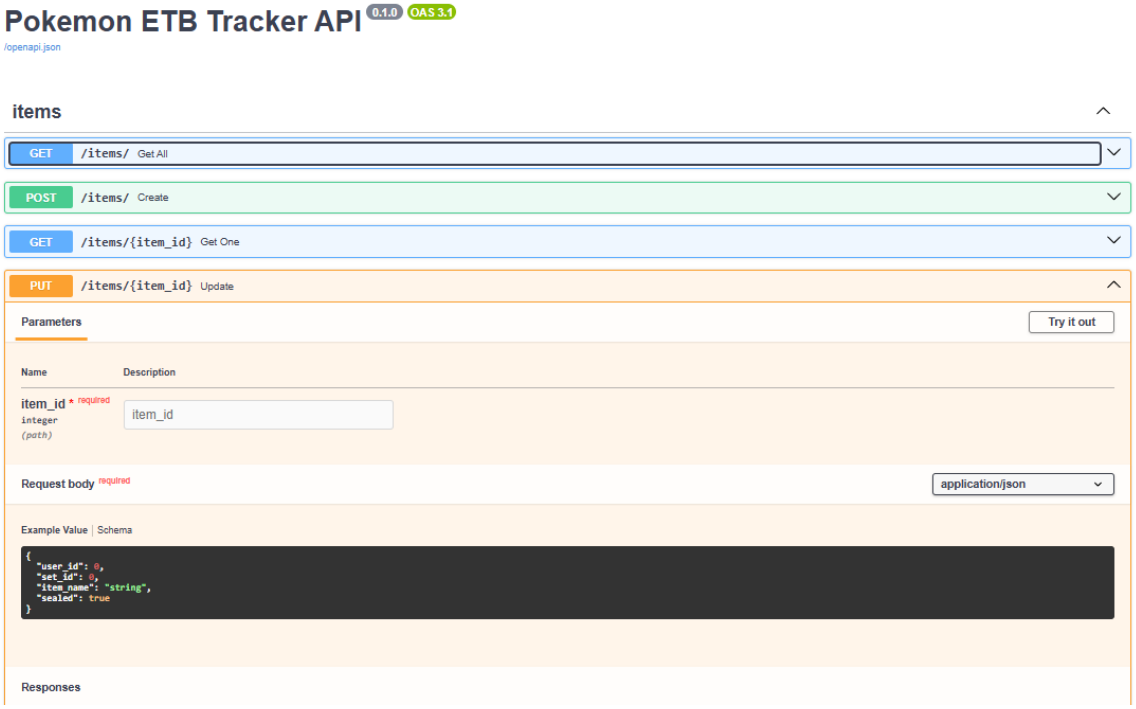
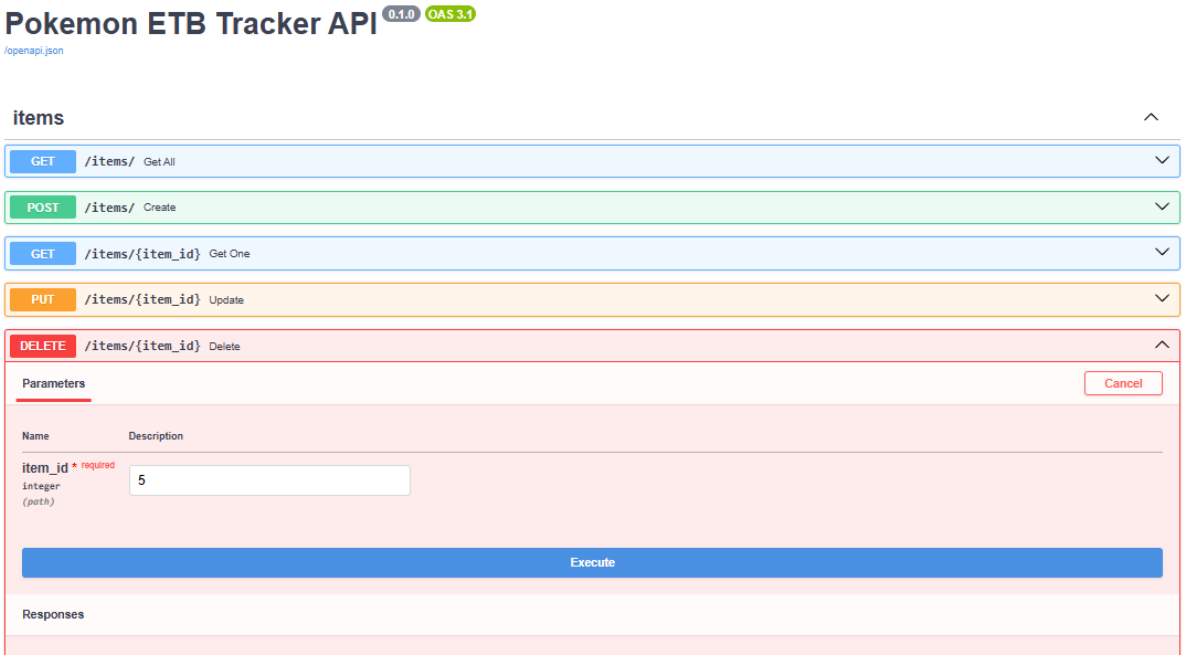
Figure 6 – Delete items